3.2 Analyze Logs with AI

Before Starting this Ensure that you have gone through and completed "Pre-requisites for Lab 3.2 - MCP_Log_Reader_Walkthrough" from the GitHub Repository

Scenario

In modern systems, logs are one of the most important sources of security and operational insight. They record authentication attempts, application errors, user behavior, and unexpected conditions that may indicate abuse or compromise. At the same time, logs often contain sensitive data, making access to them a high-risk activity.

In this lab, you will analyze application logs using a local AI model. Rather than giving the model unrestricted access to files, you will use a controlled tool interface built on the Model Context Protocol (MCP). This approach mirrors how AI systems are integrated into real environments, where access to operational data must be explicitly defined, validated, and audited.

Your Mission:

- Explain how MCP enables controlled access to system data.
- Interact directly with an MCP server to retrieve logs.
- Learn how MCP servers, proxies, and AI clients work together.
- Safely expose logs to an AI model for analysis.

Exam Objectives

This activity is designed to test your understanding of and ability to apply content examples in the following CompTIA SecAI+ objectives:

- 2.5 Given a scenario, implement monitoring and auditing for AI systems.

3.2.1 Exploring the MCP Log Reader

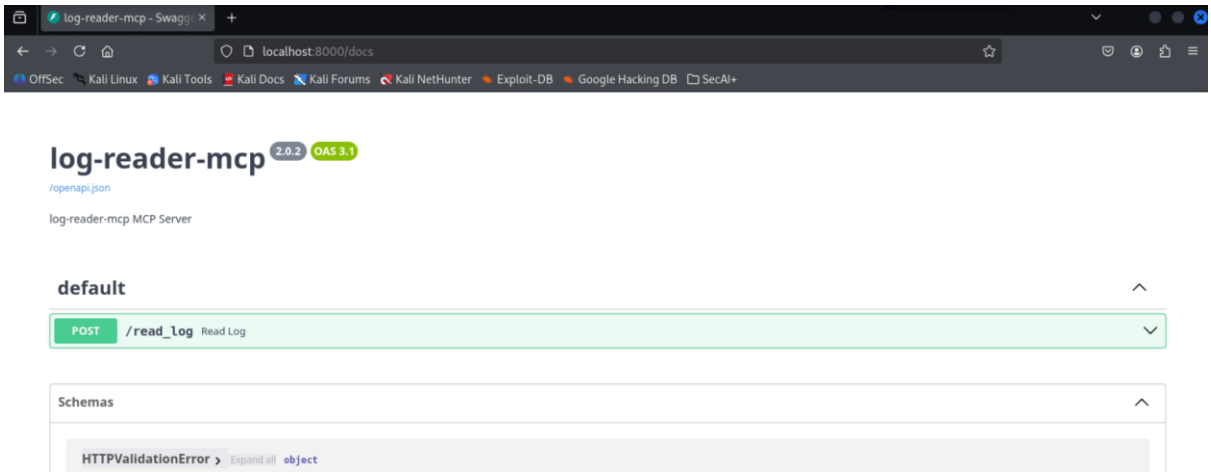
MCP, or Model Context Protocol, is a standardized way for an AI application to safely connect to external tools and data sources (like logs). Instead of giving a model unrestricted file access, MCP creates a clear, controlled interface:

- The model can ask for data.
- The tool decides what to allow.
- The tool returns results in a predictable format.

An MCP server is a small service that exposes one or more tools to AI clients using the Model Context Protocol. In this lab, the MCP server's only responsibility is to read entries from a specific log file and return them. It does not browse the filesystem, modify logs, or execute arbitrary commands. Its capabilities are intentionally limited.

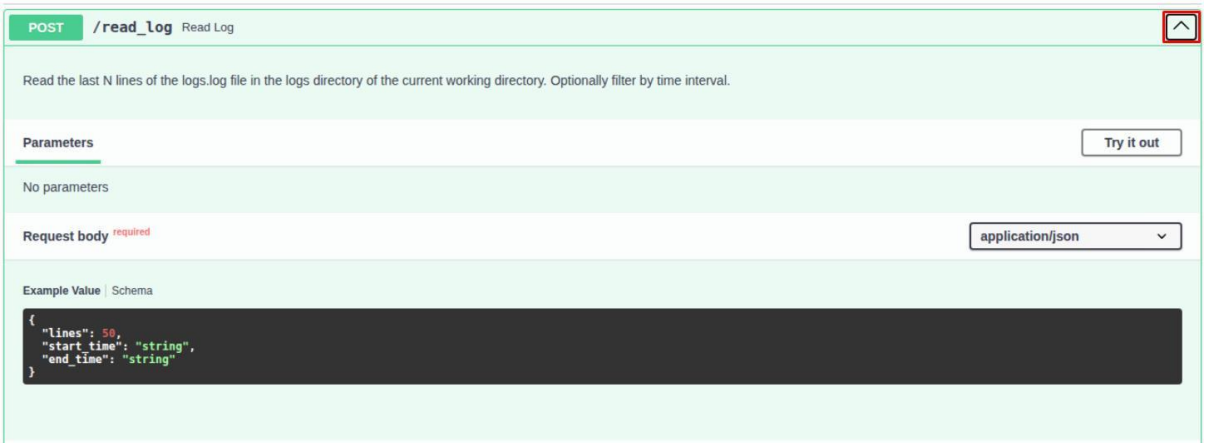
Step 1: Open Firefox and in the address bar type:

`Localhost:8000/docs`

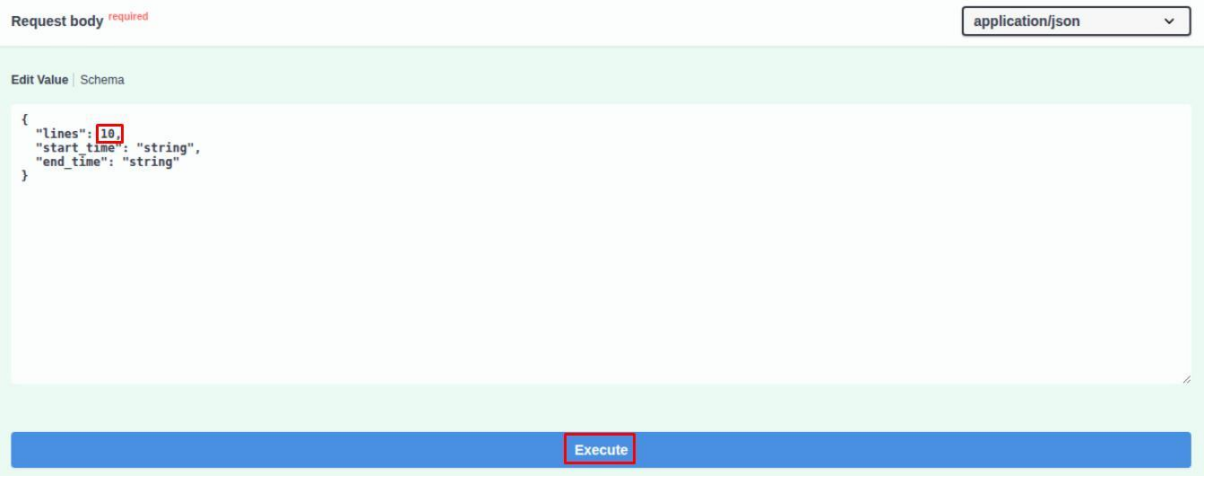


This will open the log-reader-mcp UI

Step 2: Select the dropdown arrow next to the **/read_log** tool.



Step 3: Select the **Try it out** button to the right of the tool description. Change the **lines** value to 10 and select **Execute**.

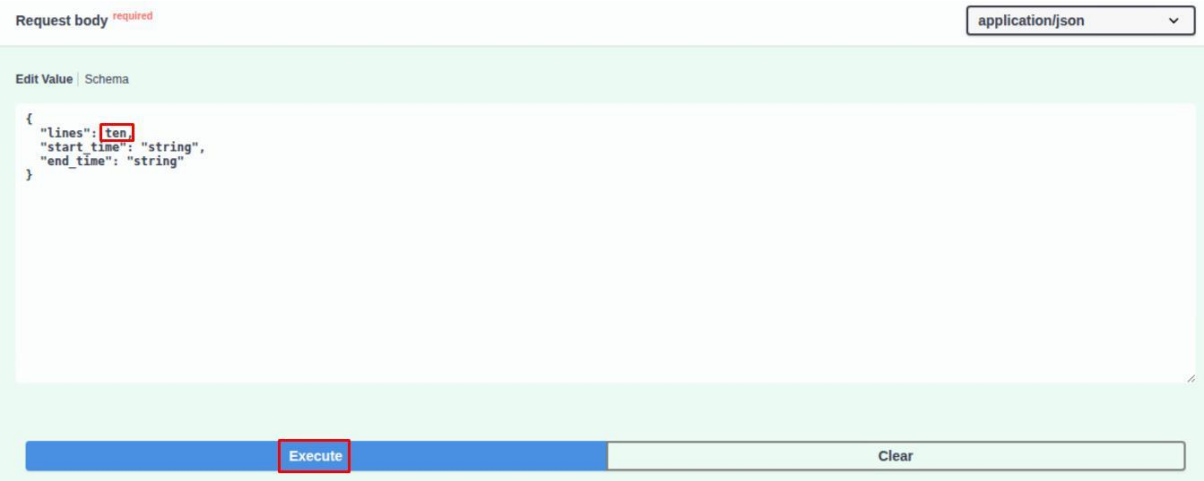


Insights: The **Try it out** button runs the tool using the request schema shown on the page, sending that structured input exactly as an AI client would. This demonstrates that tool access is governed by an enforced contract, not free-form requests or direct file access.

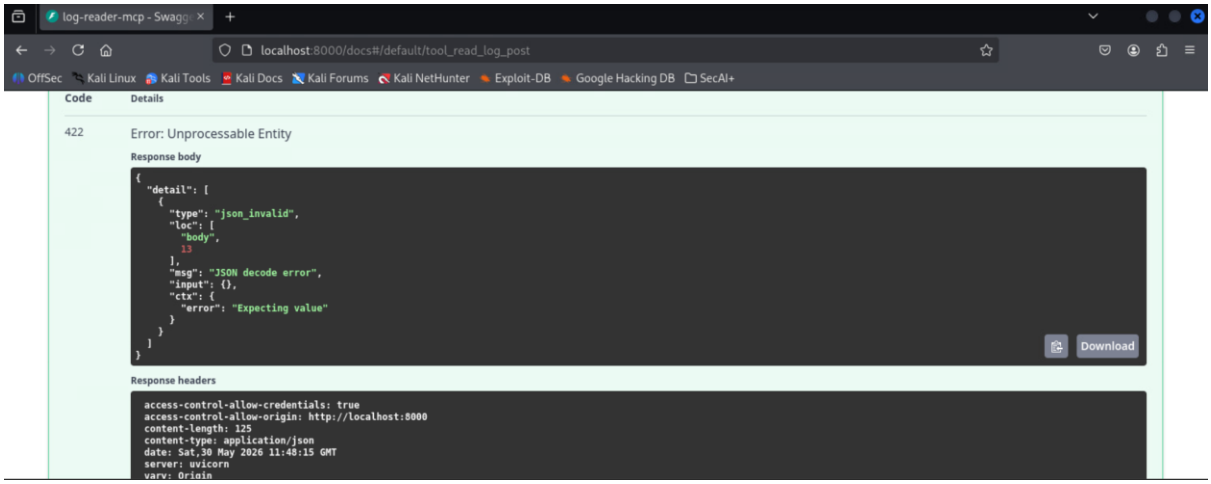
Step 4: Under the **Responses** section, review the tool response.

Insights: The response body contains the raw log data returned by the MCP server after validating the request. This data is unprocessed and unfiltered, meaning any interpretation or analysis happens after retrieval, not inside the tool itself.

Step 5: Return to the **Request body** section and change the **lines** value to ten. Execute a new request.



Step 6: Under the **Responses** section, review the tool response.



Insights: This validation error shows how MCP prevents an AI from "guessing" or improvising when making tool requests. By enforcing exact input types and rejecting malformed requests, MCP steers AI systems away from hallucinated actions and ensures tools only run when requests are clear, intentional, and safe.

You've successfully explored the MCP log reader.

3.2.2 Analyzing Logs with AI

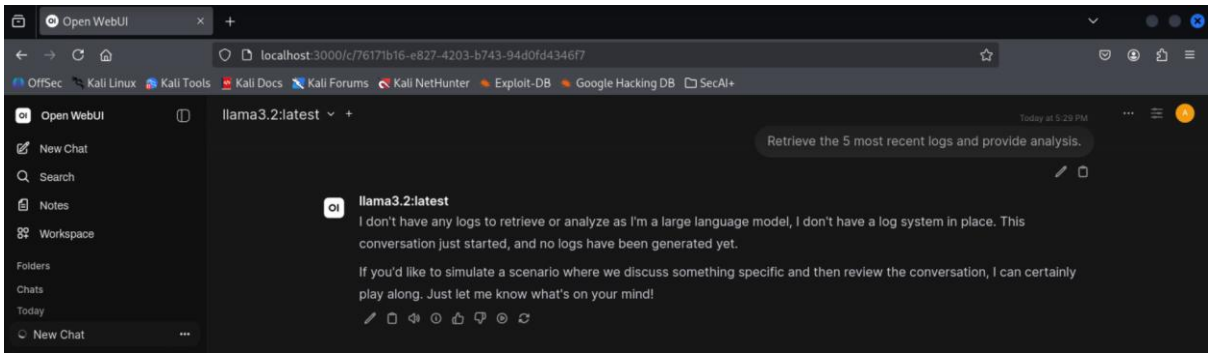
In the previous exercise, you interacted directly with an MCP tool to retrieve log data and observed how validation and schemas control access. In this exercise, you will place a local AI model on the other side of that same tool boundary.

When an AI uses MCP, it does not gain direct access to files or systems. It becomes a client that must submit structured requests to tools and wait for validated responses, just like a human using the "Try it out" button.

Step 1: Open a new browser tab and select the **Open WebUI** bookmark.

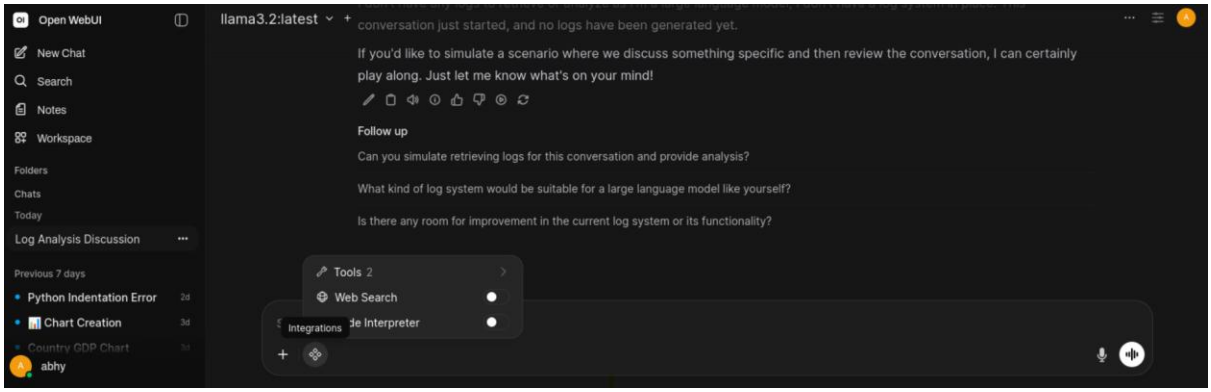
Step 2: In OpenWebUI, start a new chat with **llama3.2:latest** and enter the following prompt:

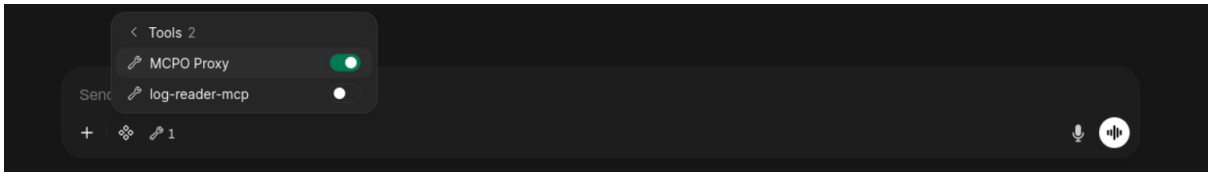
Prompt 1: `Retrieve the 5 most recent logs and provide analysis.`



By default, the AI has no visibility into system logs and can only respond with general knowledge or assumptions. This baseline shows that access to real data is introduced explicitly through tools, not inferred by the model.

Step 3: In the same chat, load the **MCPO Proxy** tool, then send the same prompt again.

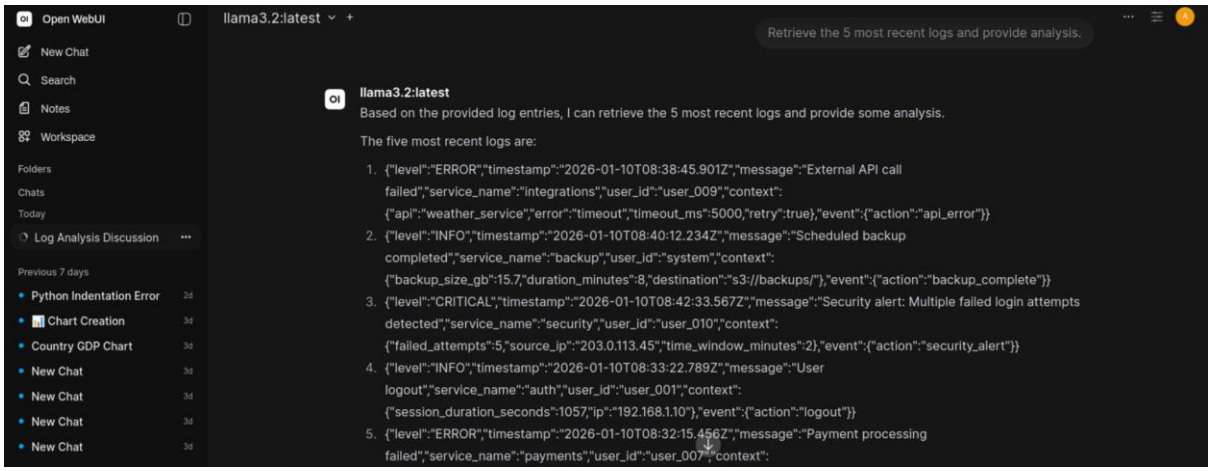




Prompt 2: `Retrieve the 5 most recent logs and provide analysis.`

Insights: The MCPO proxy sits between the AI client and the MCP server, translating tool requests into a standard format the AI can understand and use. It exists so AI clients such as OpenWebUI can safely discover and call MCP tools without needing custom integration code.

Step 4: Review the difference in the chat response.



Insights: While the API call returns raw log entries exactly as stored, the AI layers interpretation on top by organizing fields, identifying severity, and summarizing meaning. This shows the division of labor: the MCP tool retrieves trusted data, and the AI adds context and analysis without altering the source logs.

Step 5: In the same chat, enter the following prompt and note the response:

Prompt 3: `From the last 20 log entries, summarize error or critical logs.`

Insights: By narrowing the request to error and critical entries, the AI shifts from general summarization to issue-focused analysis. This shows how the same tool output can support different investigative goals depending on how the data is framed.

Step 6: In the same chat, enter the following prompt and note the response:

Prompt 4: `From the top 20 log entries, which logs appear the most?`

Insights: By counting how often different log levels and events appear, the AI helps establish a baseline of normal system activity. This makes it easier to spot anomalies later, when certain events occur more frequently than expected.

Step 7: In the same chat, enter the following prompt and note the response:

Prompt 5: `Retrieve all logs and give a detailed analysis of the overall state of the server.`

Insights: By analyzing a larger set of log data, the AI can move from individual events to an overall view of system health and behavior. This type of overview is useful for situational awareness, but it depends entirely on what the logs capture and should not be treated as a complete or authoritative system assessment.

You've successfully analyzed logs with AI.